Wind Tunnel Project

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RESEARCH $\longrightarrow$ **DEFINE** $\longrightarrow$ **DESIGN** $\longrightarrow$ **BUILD** $\longrightarrow$ **VALIDATE**

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Research Status

Konstantin

- Airfoil design
- Measurement locations on the airfoil

Bas

- XFOIL simulations
- C_p measurement methods
- Methods for measuring C_p over x/c

Dante

- Open-loop vs. closed-loop wind tunnels
- Advantages and disadvantages of both

Robbe

- Communication protocols
- Controller/ESP to Raspberry Pi communication

Research results will be used to define the final measurement setup and system architecture.

Where Are We & Where Are We Going?

Where are we?

- Existing open-loop wind tunnel available
- Initial research performed
- Airfoil measurement methods investigated
- XFOIL as theoretical reference
- Communication options investigated



Where are we going?

- Final measurement concept
- Final airfoil design
- Sensor and hardware selection
- Control and communication system
- Complete integrated wind tunnel
- Experimental validation against XFOIL

From research $\longrightarrow$ to engineering decisions

What Has to Be Done?

Airfoil & Measurements

- Minimum pressure-point count
- Optimal x/c locations
- Literature and XFOIL basis
- Final airfoil and pressure taps

Sensors

- Final pressure sensors
- Final airspeed sensor
- Required range and accuracy
- Required sampling rate

Electronics & Communication

- Measurement architecture
- Controller/ESP to Pi protocol
- Electronics and interfaces

Control

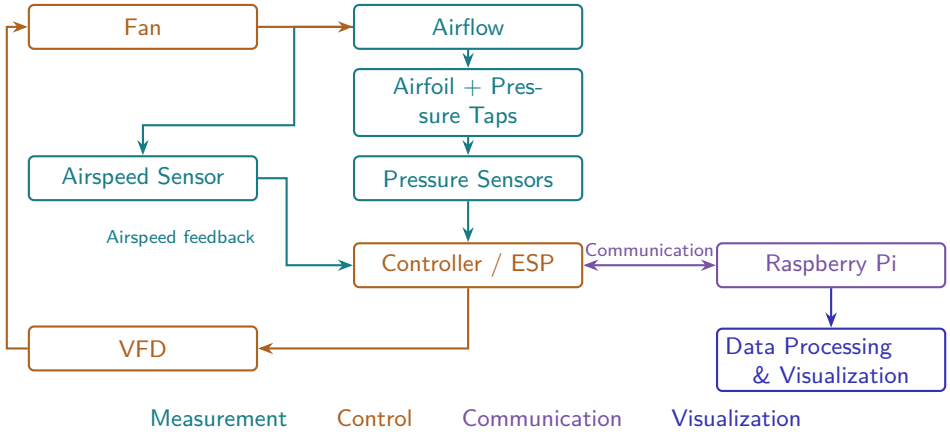
- Develop airspeed control
- Integrate fan/VFD control
- Define feedback measurement

Validation

- Calibrate sensors and quantify uncertainty
- Measure $C_p(x/c)$ and compare with XFOIL
- Define final wind tunnel acceptance criteria

System Architecture

Proposed architecture: hardware and protocol to be selected



Planning & Next Steps

Phase 1	Phase 2	Phase 3	Phase 4
DEFINE	**DESIGN**	**BUILD & INTEGRATE**	**VALIDATE**
• Measurement points • Airfoil • Sensors • Communication • Requirements	• Mechanical design • Electronics • PCB/wiring • Software architecture • Control system	• Manufacture airfoil • Install sensors • Implement software • Integrate VFD/fan control • Integrate complete system	• Calibration • Wind tunnel testing • C_p measurements • XFOIL comparison • Accuracy evaluation

DEFINE $\longrightarrow$ DESIGN $\longrightarrow$ BUILD & INTEGRATE $\longrightarrow$ VALIDATE

Concrete Next Steps

Action	Deliverable
1. Define requirements	Requirements list: airspeed, measurement range, accuracy, sampling rate and acceptance criteria.
2. Define the measurement setup	Pressure-point count and x/c locations, justified by literature and XFOIL.
3. Select components	Sensor, electronics and communication choices, checked against the requirements.
4. Complete the design	Airfoil drawing, wiring diagram, software architecture and control loop.
5. Prepare the test plan	Calibration procedure, uncertainty assessment and C_p comparison with XFOIL.

Requirements $\longrightarrow$ Decisions $\longrightarrow$ Design $\longrightarrow$ Test plan